L: speaker script for the PALS innoWAH! pitch (embedded-systems club, 23 Oct 2026)

Deck: `L_pitch_final.pptx` (31 slides). Speaker: Navin (team L Lawliet, CKCET). Core talk about 9 minutes; a 5-minute cut is in section 6. Every number is said with its label: **measured**, **simulated** or **estimated**.

1. The one sentence

> "AI stops learning the day it ships. L is a small chip design that keeps learning on the device after it ships, without forgetting what it knew, using only counting: no multiplier, no matrix, no backpropagation. And I will show how real AI models can be altered, or newly designed, to run on it."

If you only get 20 seconds, say that, then: "Everything I show is simulation, one real GPU measurement, and design work. Nothing has run on a board yet, and I say exactly what that means."

2. The three questions every judge will ask, and your answers

Memorise these. They are slides 3, 17, 22 and 23. How you tested it is slide 7. Where the GPU uses more energy and L less is slide 16.

Q1. "What is wrong with the transformer and the GPU? Why change anything?" (slide 3) "Nothing is wrong with them for what they were built for: training and big batches in a data centre. Three things hurt on a device. 1. **The model is frozen.** It is trained once, then fixed. New data means retraining in a data centre, and a network trained on new things forgets old ones. 2. **Every token reads everything.** Making each word reads every weight, and attention keeps a memory that grows with the text. At one input at a time the chip waits for memory, not for maths. 3. **A GPU needs a crowd.** I measured it on my RTX 3050, twice: one decision costs 12.6 to 69 millijoules alone and 0.29 to 1.95 in a batch of a thousand, 35 to 43 times worse when alone. A phone or a sensor decides one at a time. The heat, cooling water and hardware turnover follow from that power. That is our motivation; I have not measured it."

Q2. "Why should we agree with you and use your chip and model?" (slide 17) "For one specific job, I can show it is better: learning on the device, with a small memory, one input at a time. In staged learning on 59 thousand images the chip ends at 95.5% and a real CNN, which I ran on my own GPU, ends at 19.6% (chip simulated, CNN measured). It needs 84 KB and no multiplier, and about 74 times less energy per decision by operation count (estimated; the board run will measure it). The design matches the model bit for bit in 41 of 41 checks. **When everything is learned at once the CNN is still ahead, 99.1 to 97.9, and I say that. We do not replace the GPU: keep it for training and big models.**"

Q3. "Why do you need to alter AI models, or make your own? What is wrong with the current architecture?" (slide 22) "The transformer is not wrong; it is built for a data centre. For a device its structure hurts in four ways: every word does dense maths on every weight; attention keeps a memory that grows with the text; the weights are frozen after training; and

learning needs backpropagation, which needs gradients and a lot of memory. **A faster chip for that same model is a GPU or a TPU, and those already exist with far more money behind them.** So the waste is in the model, and the model must change with the chip: a constant-size state, sparse activity so only a few rows are touched, low-bit or counting weights, and local learning by counting. We either alter an existing model to those rules or design a new one, and build the chip for exactly that. Model and chip designed together is the L bet. The small version works, as classifiers. A first test of a counting language model is in: it learned English text online in one pass, and its quality grew with memory, 4.8 bits per character at 8 KB to 1.8 at 8 MB (simulated algorithm test, not on the chip, not yet compared with a neural model). It has tests that can prove it wrong."

Q4. "Can real AI models run on your chip?" (slide 23) "Three routes, and I will be exact about the status. 1. **Alter an existing model.** Ternary models, with weights of minus one, zero and plus one, turn multiplications into additions; published work runs language models of billions of parameters this way. That still has a matrix of additions, so it breaks our no-matrix rule, but it shows the multiplier is not needed. 2. **Design our own model.** A counting language model: hash the recent words into a sparse code, a table of counters predicts the next word one bit at a time. The counting classifier works in simulation. A first test of the language model is in: it learned English text online in one pass and quality grew with memory, 4.8 bits per character at 8 KB to 1.8 at 8 MB (simulated algorithm test; not yet on the chip, not yet compared with a neural model). 3. **Grow the memory.** Big tables go in outside DRAM; only a few rows are touched per word, so bandwidth stays small. First we test, in simulation, whether quality grows with table size. **We claim the route and the tests that can prove it wrong, not a billion-parameter result.** Today's chip is the small version of the idea."

If they push: "Is it proven at billions of parameters on your chip?" → "No. It is a hypothesis with falsification tests: it fails if quality drops at larger sizes without the matrix, if a fused GPU implementation gets within about 3 times, or if a transformer chip with the same memory matches it."

Q5. "How did you test it? Did you run real AI models?" (slide 7) "Three ways. Standard tasks on an integer model of the chip, which is exactly its arithmetic: digits, clothing images, spoken digits and words, all at once, in stages, with noise, with two senses, with a new class from one example, with two chips merged; 3 to 5 seeds, a held-out speaker. A real CNN and a small network on the same data, the CNN on my own RTX 3050 so time and energy are measured. And the design itself: the Verilog matches the model bit for bit in 41 of 41 checks, the real-cell netlist matches, a formal equivalence check passed. **I did not run a language model or any large model on the chip. The models I ran are classifiers; the CNN is only the ruler. A board run is the next step.**"

Q6. "For which process does the GPU use more power and L less?" (slide 16) "Four processes, with labels. One image alone, picture to answer: GPU 12.6 to 69 mJ in my two runs (measured) against about 14 microjoules for the chip (layout estimate with switching measured in simulation; the encoder is 13 of the 14). One image in a crowd of 1,024: GPU 0.29 to 1.95 mJ (measured); a device decides one at a time. Learning from 59,063 digit images: the CNN's four-pass training used 1,148 joules (measured), the chip learns once, in place, about 4 joules (estimated, older gating assumptions with the encoder added). Learning a new class later: the GPU must retrain, and in stages it forgets; L learns in place. **Orders of magnitude, not a final**

ratio: the chip numbers are estimates, the GPU numbers are the whole GPU and changed between my two runs; the board run will measure L. The GPU still wins at training big models and batched throughput, and end to end the FPGA build is not faster than this GPU."

3. How to pitch, and how not to

Do - Order: problem (devices cannot learn), what is wrong with today's models, the idea, the proof and how it was tested, why use us, why the model must change, how bigger models can run, limits, ask. - Say the label with every number. Judges trust the label more than the number. - Say what is weaker before they find it (slide 25). - Frame it as an **embedded system**: 84 KB, no multiplier, a small cheap FPGA, a UART interface, learning in the field. - Slow down on: **95.5 against 19.6**, **84 KB**, **35 to 43 times worse at batch 1**, **41 of 41**. - Not sure? "I have not measured that. Here is what I did measure."

Do not - Do not say "replaces the GPU", "faster than a GPU" or "runs ChatGPT-size models". It does not, today. - Do not say "never forgets". Say "keeps old classes far better than a normal network when learning in stages". - Do not say "fabricated", "taped out", "tested on silicon", "runs on the Tang Nano". - Do not call the energy number measured. The 74 times is an operation count (a lower bound) and the 14 µJ is a layout estimate; only the RTX 3050 numbers are measured. Do not say the chip is faster than the GPU: end to end at 27 MHz it is not (0.83 ms against 0.42 ms at batch 1); say its advantage is energy and learning on the device. - Do not say "we solved catastrophic forgetting". Say "greatly reduced, in our tests". - Do not claim heat, water or e-waste savings as results. They are motivation. - Do not read slides. One idea per slide.

4. What you can claim, and what you cannot

You can say (with the label)	Source / label
Learning digits in 5 stages: chip about **95%**, a normal network about **19.5%**	SIMULATED chip
59,063 digit images in stages: chip 95.5%, CNN 19.6%; all at once: CNN 99.09%, chip 97.91%	chip SIMULATED, CNN **MEASURED** on my RTX 3050 laptop GPU
One image costs 12.6 to 69 mJ at batch 1 and 0.29 to 1.95 mJ at batch 1024 on my GPU (two runs; 35 to 43 times worse alone)	**MEASURED** (NVML)
Memory for images **84 KB**, no multiplier; fits a Tang Nano 20K in place and route (89% logic, 0 multipliers)	design fact / SIMULATED place and route
RTL bit-exact with the model in **41 of 41** checks; core netlist in SKY130 cells matches the model; post-layout netlist proved equivalent to the synthesised one	SIMULATED
Picture to answer: the encoder takes 21,378 cycles and the decision about 1,500 (FPGA build) or 320 (pipelined), so about 0.83 ms at	chip cycles SIMULATED, times ESTIMATED; GPU **MEASURED**

You can say (with the label)	Source / label
the 27 MHz FPGA clock and 0.45 ms at 50 MHz; the decision alone is 55.7 µs or 11.7 µs. GPU batch 1: 0.42 to 1.8 ms	
Picture to answer energy: about 14 µJ, about 30 mW while encoding (layout power analysis with switching measured in simulation on real MNIST images); GPU 12.6 to 69 mJ, about 30 to 40 W while running	chip ESTIMATED; GPU **MEASURED**
Picture to answer, chip against GPU: about 20 to 5,000 times less energy depending on the GPU's power state and batch	ESTIMATED against MEASURED
About 74 times less energy per decision than an 8-bit network, counting operations	ESTIMATED
Ternary language models of billions of parameters are published	published research; verify before quoting
A counting classifier works with counts only	SIMULATED
A counting language model, one pass online on 561 KB of English text: 4.83 bits per character at 8 KB, 1.80 at 8 MB (xz offline 1.54)	SIMULATED algorithm test; not on the chip; no neural baseline yet
Two chips merge by adding counters (91.4% on ten digits after one night, board-demo emulator); a new class from one example: digits 54 to 70, clothing 50 to 64	SIMULATED

Do NOT claim	Why
It runs on hardware, a board or a chip	Not done; the board run is step 1 of the ask
It beats or replaces GPUs	Different job; a CNN is ahead when everything is learned at once
A billion-parameter model runs on L	A hypothesis; tests not yet run
Measured chip power or energy	Only an estimate until the board is read
"Never forgets" / "solved forgetting"	A rotated-digits test dropped old digits from 95 to 45% without the frozen-memory option
"First chip that learns continuously"	Hyperdimensional-computing accelerators and Loihi 2 exist; what is new is the combination
Heat, water, e-waste savings	Motivation, not measured
Large vocabularies, real audio-visual data, real noise	Small tests only

5. The script, slide by slide (31 slides)

Times are for the 9-minute talk. Say the bold sentence even if you skip the rest.

1. Cover (0:15). "We are L Lawliet from CKCET and the project is called L. **Every number is labelled simulated, estimated or measured, and nothing has run on a real board yet.**"

2. AI stops learning the day it ships (0:35). "A phone, a camera, a sensor node: the AI inside was trained once on a big machine. If the world changes, it does not learn. **I want the device itself to learn, without forgetting.**"

3. What today's model and GPU get wrong (0:60). Answer Q1 above, three points, then: "**Heat, water and hardware turnover follow from the power: that is our motivation, not a result.**"

4. The idea: a brain, as a chip (0:40). "A brain counts, compares and sleeps. L does the same: a table of small counters decides, and at night the chip dreams about old classes from its own counters, so new learning does not overwrite them. **No multiplier, no matrix, no backpropagation.**"

5. Learns in stages and keeps what it knew (0:45). "Five stages, two digits at a time. A normal network ends near 19.5%, because it forgets. Our chip ends near 95% (simulated). **On 59,000 images a real CNN on my GPU ended at 19.6% in stages and the chip at 95.5%.**"

6. Close to a CNN trained on everything (0:35). "When everything is learned at once the CNN wins, and I say so: 99.09 against 97.91 on digits, about 2 to 3 points on clothing. **Close, without any multiplier.**"

7. How we tested it (0:45). "Three ways. One: standard tasks on an integer model of the chip, which is exactly the arithmetic it does: handwritten digits, clothing images, spoken digits and words; all at once, in stages, with noise, with two senses, a new class from one example, two chips merged; 3 to 5 seeds and a held-out speaker. Two: a CNN and a small network trained on the same data, with the CNN run on my own RTX 3050 so accuracy, time and energy are real. Three: the Verilog matches the Python model bit for bit in 41 of 41 checks, the real-cell netlist matches, and a formal equivalence check passed. **What we did not test: a language model or any large model on the chip, a board run, real-world noise. The models we ran are classifiers; the CNN is only our ruler.**"

8 to 12. Scale, sound, two senses, new class, sharing (1:15 total; pick two). "It still works at 60,000 images with one rule that sets its sleep interval. It hears spoken words and noise hurts less (simulated). Two senses help each other. A new class from one example after shipping. Two chips merge by adding counters: a normal network cannot do that."

13. The design in one picture (0:45). "Camera and microphone, a retina and a cochlea that make on-off bits, a sparse code, the fast table that decides, the dream memory, the slow table, a day-night sequencer, and a UART. **Counts, compares, shifts, table lookups. That is all.**"

14. Small, cheap, no multiplier (0:35). "84 KB. It fits a 25-dollar Tang Nano 20K with zero multipliers (simulated place and route). About 74 times less energy than an 8-bit network, counting operations. **That 74 times is an estimate; I will measure it on the board.**"

15. Picture to answer: chip against a real GPU (0:55). "Same job on my RTX 3050, picture to answer. Measured, in two runs: about 0.4 to 1.8 milliseconds and 12 to 69 millijoules at batch one; up to 98,000 images a second at batch 1024. The chip design: the encoder takes 21,378 cycles and the decision about 1,500, so about 0.83 milliseconds at the 27 megahertz FPGA clock and about 14 microjoules, about 30 milliwatts while encoding, from the layout with switching measured in simulation. **So end to end the FPGA build is not faster than this GPU; its advantage is energy, about 20 to 5,000 times lower depending on the GPU's power state, and learning on the device. Chip numbers are estimates, not a board run.**"

16. Where the GPU uses more energy, and L less (0:60). Say the four processes from the table on the slide. "One image alone: GPU 12.6 to 69 millijoules, measured; the chip about 14 microjoules, estimated. In a crowd of 1,024 the GPU improves to 0.29 to 1.95 millijoules, but a device decides one at a time. Learning from 59 thousand images: the CNN used 1,148 joules over four passes, the chip learns each example once, in place, about 4 joules, estimated. A new class later: the GPU retrains and in stages forgets, the chip learns in place. **Orders of magnitude, not a final ratio. The GPU still wins at training big models and batched throughput. The board run will measure L.**"

17. Why use L instead, for this job (0:50). Answer Q2 above. End on the green box: "**Use L when the job is learning on the device, with a small memory, one input at a time. Keep the GPU for training and big models.**"

18. Why this is an embedded system (0:35). "Tight memory, no multiplier, a small FPGA, a camera-style input and a UART. It learns in the field, sleeps when idle, and its memory can be moved between devices."

19. Verified bit for bit (0:30). "41 of 41 checks: the RTL matches the Python model exactly, and the layout flow in the open tool chain reaches routing with zero violations by the router's own count. **That is a layout in simulation, not a fabricated chip.**"

20. The design, so you can verify it (0:15). "Everything is in a public repository: the model, the RTL, the checks."

21. What L solves, and what it does not (0:30). "It solves a small continual learner for sensors. It does not run existing models as they are, and it is not a GPU."

22. Why the model must change, not just the chip (0:60). Answer Q3 above. "A transformer is not wrong; it is built for a data centre. For a device its structure hurts: dense maths on every weight, attention memory that grows, frozen weights, learning that needs backpropagation. A faster chip for it is just a GPU or a TPU. **So the model has to change with the chip: that is the L bet.**"

23. How real AI models can run on L (0:70). Answer Q4 above, three routes. End on: "**We claim the route and the tests that can prove it wrong, not a billion-parameter result.**"

24. The hypothesis we will test (0:30). "A brain-style model with constant-size state and sparse activity, on a chip built for it, needs less memory traffic and energy per token than a transformer on a GPU at the same quality. It is a hypothesis. It fails if quality drops at larger sizes, if a fused GPU implementation gets within about 3 times, or if a transformer chip with the same memory matches it."

25. Where it is level, and where it is weaker (0:50, do not skip). "Level: digits and short spoken-word tests against a CNN. Weaker: clothing in stages is about 82%, five points under its one-pass number; a trained CNN is about two points ahead on clothing. Tests are small, noise is simulated, the two-sense data is paired by class. **A few steps still run on the host.**"

26. Not the first (0:25). "Hyperdimensional-computing accelerators and Intel Loihi 2 exist. What is new is the combination: dreams drawn from small counters, no multiplier, merging by adding counts."

27 to 29. What comes next, two tracks, silicon plan (0:40 total). "Next is the board run. The second track is the model-running chip for the bigger problem. We write the design independent of the node. A 5-nanometre-class chip costs tens of millions of dollars and we do not claim it."

30. What we need to go further (0:50). "Step 1: a Tang Nano 20K, about 4,200 rupees, for the first measured numbers. Pass: bit-exact with simulation, accuracy and power measured. Step 2: a micro-chip on a shared shuttle, about 100 to 300 dollars a tile. Step 3: a mature-node chip through a university or programme partner. **Today I ask for step 1, and for a mentor or partner for step 2.**"

31. Close (0:15). "AI that keeps learning, on the device, with counting, and a clear route to bigger models, with tests that can prove us wrong. Thank you."

6. The 5-minute cut

Slides 1, 2, 3, 4, 5, 6, 7, 13, 15, 16, 17, 22, 23, 25, 30, 31. Say the bold sentences only. Skip 8 to 12, 14, 18 to 21, 24, 26 to 29.

7. More hard questions, short answers

- **Has it run on hardware?** "No. The RTL is bit-exact with the model and fits the Tang Nano in place and route (simulated). The board run is step 1 of my ask."
- **Does it match a neural network?** "Close, not equal: 97.9 against 99.1 on digits when everything is learned at once. In stages it is far ahead, because a normal network forgets (95.5 against 19.6)."
- **Why not fine-tune a transformer on the device?** "Fine-tuning needs gradients and a lot of memory, and a fine-tuned network forgets. L learns by counting, in place, with dreams that protect old knowledge."
- **Why not just use a bigger GPU?** "For training and big models, you should. L is for a device that must learn after shipping within a tiny power and memory budget, with no server."
- **Is the energy number real?** "The 74 times is an operation count, an estimate. The GPU numbers are measured. The chip's energy gets measured on the board."
- **Is a counting language model any good?** "It will start at roughly n-gram quality, which is weaker than a transformer. The experiment is to see whether dreams and richer codes close the gap, and whether quality grows with memory."

- **What are dreams?** “While idle, the chip draws pretend examples of old classes from its own counters and learns them again, so new learning does not overwrite old knowledge. It stops by itself when surprises become rare.”
- **Is this like a real brain?** “Brain-inspired, not brain-identical. Local counting, sleep and replay are brain-like; the sense layer and the dream sampler are engineered.”
- **Where did it fail?** “Staged clothing at scale, about 82%. Cheap fixes failed. A multi-prototype option helps by about 3 points but costs memory and is not in the RTL yet.”
- **What will you do with money or time?** “Board run first. Then the counting language model experiment, better dreams for clothing, a write-up and a demo video.”
- **Is the data realistic?** “Standard digit and clothing sets and a small spoken-word set. Larger vocabularies and real noise are not tested.”

8. Before you go on stage

- Open the deck at slide 1; check slides 3, 7, 15, 16, 17, 22, 23 and 30 render.
- Know the numbers cold: **95.5 against 19.6**, **35 to 43 times worse at batch 1**, **84 KB**, **41 of 41**, **about 14 µJ against 12.6 to 69 mJ**.
- If a slide runs long, drop slides 8 to 12 first.